Repeated Identical In-District Early-Vote Pairs That Overwhelm the Random-Chance Hypothesis

Junho Kim
No affiliation
e-mail: junhokim0331@gmail.com

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Abstract

This paper examines whether the repeated identical vote-count pairs observed in in-district early-vote counting units after the June 3, 2026 Korean local election can be explained by ordinary chance. The starting point is the public report of identical vote counts in twelve places nationwide, including ten Gwangju-Jeonnam rows where the same two leading candidates received identical counts across paired counting units. This study rechecks the twelve event rows by extracting them directly from the National Election Commission election-statistics system's official counting-unit HTML pages. The Gwangju-Jeonnam case is defined as five repeated identical vote pairs within the same election, the same top-two candidate combination, and the same in-district early-vote category.

To build a historical baseline, this study parses 81,701 early-vote counting-unit rows from official National Election Commission and public-data files from 2014 through 2025. The primary benchmark is the actual top-two candidate vote-pair distribution in governor races. Across 51 governor constituency contests and approximately 1,514,172 actual comparison pairs, only 15 identical top-two vote pairs are observed, and the maximum repeated-pair count within any one constituency contest is three. Under the expanded benchmark, the probability of observing at least five repeated pairs in the 2026 Gwangju-Jeonnam setting is approximately 0.115% using a Poisson approximation with $N = 393$ and $K = 100,944.8$. In a nonparametric resampling test drawing 393 units without replacement from the historical governor top-two pair pool, no simulation among 200,000 trials produces five or more repeated pairs; the rule-of-three 95% upper bound is approximately 0.0015%.

This paper also checks the difference between early-vote and election-day two-party vote shares in the 2016, 2020, and 2024 National Assembly elections. In 2020 and 2024, every analyzable constituency shows a higher Democratic-party two-party share in early voting than on election day. This auxiliary test is not treated as direct proof of a specific misconduct mechanism, but it reinforces the need for a broader raw-data audit. The conclusion is not a legal finding that a specific act of misconduct has been proven. It is a narrower statistical conclusion: even under conservative assumptions, the public data contain anomalies that weaken the random-chance hypothesis and require independent access to raw counting records.

Keywords: in-district early voting, identical vote pair, election integrity, duplicate probability, Poisson approximation, early-vote deviation, raw-data audit

1 Introduction

After the 2026 Korean local election, several reports noted that two leading candidates received exactly the same vote counts in different in-district early-vote counting units. A TV Chosun report dated June 8, 2026 stated that identical vote counts appeared in twelve places nationwide, including ten Gwangju-Jeonnam rows where Democratic Party candidate Min Hyung-bae and People Power Party candidate Lee Jung-hyun received the same paired vote counts. The same report also presented the Incheon Songdo 1-dong and Songdo 2-dong case, where Democratic Party candidate Park Chan-dae received 3,030 votes and People Power Party candidate Yoo Jeong-bok received 1,440 votes in both units.

The subject of this paper is statistical rather than partisan. The questions are:

1. How often has a pattern of five or more identical top-two candidate vote pairs occurred in the same election, same candidate combination, and same in-district early-vote category in historical official data? 2. Under a simple duplicate-probability model, how likely is the observed repetition? 3. Can the observed pattern be closed as ordinary random coincidence, or does it constitute a statistical anomaly requiring raw-data verification?

This paper answers the third question cautiously. Statistical improbability is not a sufficient condition for proving misconduct. However, when the improbability is strong enough, an election-management institution cannot reasonably end the matter by merely saying ”coincidence.” It must produce the raw records that can either explain or falsify the anomaly.

1.1 Research Hypotheses

The primary test concerns repeated identical vote pairs.

- Null hypothesis H_0: The five identical top-two candidate vote pairs observed in the 2026 Gwangju-Jeonnam in-district early-vote results are compatible with the duplicate-pair structure observed in past governor elections. In plain terms, the event is something that historical data would make plausible.
- Alternative hypothesis H_1: The five identical top-two candidate vote pairs in the 2026 Gwangju-Jeonnam in-district early-vote results are difficult to reconcile with the historical empirical distribution. In plain terms, the event is an outlier relative to past official data.

The secondary test concerns the difference between early voting and election-day voting.

- Null hypothesis H_0': The vote-share difference between early voting and election-day voting is explainable within ordinary constituency-level voter self-selection and voting-behavior differences.
- Alternative hypothesis H_1': The early-vote and election-day vote-share difference repeats in the same direction across many constituencies and remains insufficiently explained without additional covariates or raw-data verification.

Both H_1 and H_1' are treated as statistical audit triggers, not legal judgments.

2 Definition of the Observed Event

The event analyzed in this paper is defined as follows:

- Election: the 9th nationwide local election held in 2026.
- Vote type: in-district early voting.
- Comparison unit: eup, myeon, or dong counting-unit results within the same metropolitan or provincial governor race.
- Candidate rule: the principal first- and second-place candidates in the relevant race.

- Event: two different counting units report the same first-candidate vote count and the same second-candidate vote count simultaneously.

The reported Gwangju-Jeonnam case is summarized as five pairs. This study re-extracts and checks these values from the official NEC election-statistics counting-unit pages.

Pair	Counting unit A	Counting unit B	Min Hyung-bae	Lee Jung-hyun
1	Songjeong 1-dong, Gwangsan-gu, Gwangju	Geumsan-myeon, Goheung-gun	1,401	120
2	Haui-myeon, Sinan-gun	Samil-dong, Yeosu-si	506	42
3	Iyang-myeon, Hwasun-gun	Byeongyeong-myeon, Gangjin-gun	444	46
4	Eomda-myeon, Hampyeong-gun	Bukha-myeon, Jangseong-gun	606	57
5	Nodong-myeon, Boseong-gun	Palgeum-myeon, Sinan-gun	356	42

The Incheon Songdo case is treated separately because it belongs to another contest.

Counting unit A	Counting unit B	Park Chan-dae	Yoo Jeong-bok
Songdo 1-dong	Songdo 2-dong	3,030	1,440

The paper therefore separates two levels:

- Internal Gwangju-Jeonnam event: five pairs in one contest environment.
- Nationwide reported event: six pairs across more than one contest.

The stronger statistical target is the internal Gwangju-Jeonnam event. The nationwide six-pair set is more exposed to post-search and multiple-comparison objections because it was found after scanning across contests.

3 Data and Reproduction Method

3.1 Historical Baseline Data

The study uses public National Election Commission and public-data election counting files from 2014 through 2025.

Election	Parsed rows
2014 local	25,600
2016 National Assembly	3,661
2017 presidential	3,684
2018 local	22,926
2020 National Assembly	3,681
2022 presidential	3,703
2022 local	10,942
2024 National Assembly	3,754
2025 presidential	3,750
Total	81,701

Aggregate rows, subtotal rows, shipboard/residential special votes, overseas votes, and out-of-country categories are excluded from the main comparison. In-district early voting and out-of-district early voting are separated, and the central analysis uses in-district early-vote rows.

At the time this paper was prepared, the official integrated XLSX file for the 2026 local election was not available in the same public-data format used for the historical baseline. However, the NEC election-statistics system was providing counting-unit results for the 9th nationwide local election. This study saves and parses the relevant POST-response HTML from the official "counting-unit result" pages and rechecks the twelve event rows. The procedure is implemented in `scripts/fetch_nec_2026_duplicate_cases.py`. The main outputs are:

- `outputs/nec_2026_reported_duplicate_cases.csv`
- `outputs/nec_2026_reported_duplicate_pairs.csv`

Thus the 2026 event values no longer depend only on a media screenshot. When the official integrated XLSX file or original counting statements become available, the same event definition should be re-applied.

3.2 Inclusion and Exclusion Criteria

The 2014 local election is included to increase baseline reliability. It is the first nationwide local election after the introduction of the nationwide early-voting system, and the public counting files distinguish in-district and out-of-district early voting. The 2010 local election and the 2012 presidential/National Assembly elections are excluded from the main baseline because they do not belong to the same in-district early-voting institutional regime.

This choice does not bias the result in favor of anomaly detection. Including the 2014 local election increases the number of historical duplicate pairs and therefore weakens the apparent rarity of the 2026 event. Even after this broader inclusion, the 2026 Gwangju-Jeonnam five-pair repetition exceeds the historical maximum.

3.3 Actual First- and Second-Place Candidate Rule

A naive analysis might compare the first two candidates listed in each file. That does not always match the reported meaning of "first and second place." This paper therefore uses the following procedure:

1. Group counting units by election, office type, constituency, and vote type. 2. Compute total votes for each candidate within the group. 3. Select the actual first- and second-place candidates by total votes. 4. For each eup, myeon, or dong in-district early-vote counting unit, form the ordered vote pair for these two candidates. 5. Count whether the same vote pair appears in two or more different units.

This is the rule closest to the reported event.

3.4 Mapping Claims to Verification Files

This paper separates statistically reproducible claims from administrative or legal claims that require additional raw records. Without this separation, a critic can attack the weakest part of a broad claim and dismiss the entire analysis. The reproducible claims in this package are:

1. The twelve 2026 event rows are reproduced from official NEC election-statistics HTML pages. 2. Those twelve rows form six identical vote pairs. 3. In the 51 historical governor in-district early-vote contests, the maximum repeated-pair count within a single contest is three. 4. With $N = 393$ and $K = 100,944.8$, the Poisson-tail probability for five or more repeated pairs in Gwangju-Jeonnam is approximately 0.115%. 5. In 200,000 nonparametric trials sampling 393 units without replacement from the historical governor actual-pair pool, five or more repeated pairs occur zero times. 6. In the 2020 and 2024 National Assembly elections, the Democratic candidate's early-vote two-party share is higher than the election-day two-party share in every analyzable constituency.

The unresolved causal claim is different: whether the identical pairs were produced by natural counting processes, data-entry or display errors, review/reclassification procedures, or another cause. That question requires original counting statements, ballot-sorter first-pass outputs, review-ballot allocation records, and input logs. The detailed claim-to-file mapping is provided in `evidence_matrix_ko.md`.

The completed conclusion of this study is therefore not "the cause has been proven." It is: "official page values and historical baselines are sufficient to define a statistical anomaly requiring raw-data audit."

4 Empirical Baseline

4.1 Historical Governor Elections

The closest comparison class for the 2026 Gwangju-Jeonnam case is past governor elections because the observed event occurred in a metropolitan/provincial governor race and in the in-district early-vote category.

Reanalysis of the 2014, 2018, and 2022 governor-election in-district early-vote data under the actual top-two candidate rule gives:

Quantity	Value
Governor constituency contests	51
Total comparison pairs	1,514,172
Identical vote pairs	15
Constituency contests with at least one identical pair	9
Maximum identical pairs within one contest	3
Constituency contests with five or more pairs	0

The observed historical identical pairs are:

Election	Constituency	First-place candidate	Second-place candidate	Pair	Units
2014 local	Gangwon	Choi Moon-soon	Choi Heung-jip	49, 80	Duchon-myeon, Gimsatgat-myeon
2014 local	Gangwon	Choi Moon-soon	Choi Heung-jip	57, 95	Buron-myeon, Mukho-dong
2014 local	Gyeongnam	Hong Joon-pyo	Kim Kyung-soo	293, 83	Bugok-myeon, Ungyang-myeon
2014 local	Gyeongbuk	Kim Kwan-yong	Oh Joong-gi	160, 23	Chojeon-myeon, Hyeondong-myeon
2014 local	Gyeongbuk	Kim Kwan-yong	Oh Joong-gi	186, 28	Jisan-dong, Andeok-myeon
2014 local	Gyeongbuk	Kim Kwan-yong	Oh Joong-gi	342, 43	Pyeonghwa-dong, Bongyang-myeon
2014 local	Jeonnam	Lee Nak-yon	Lee Seong-su	124, 18	Nam-myeon, Eomda-myeon
2014 local	Jeonnam	Lee Nak-yon	Lee Seong-su	190, 33	Bugil-myeon, Heuksan-myeon
2014 local	Jeonbuk	Song Ha-jin	Park Cheol-gon	145, 41	Ungpo-myeon, Boan-myeon
2014 local	Jeonbuk	Song Ha-jin	Park Cheol-gon	174, 27	Ipyeong-myeon, Ibaek-myeon
2014 local	Jeonbuk	Song Ha-jin	Park Cheol-gon	174, 50	Seosu-myeon, Jungang-dong
2014 local	Chungnam	Ahn Hee-jung	Chung Jin-suk	54, 61	Sicho-myeon, Munsan-myeon
2018 local	Daegu	Kwon Young-jin	Im Dae-yoon	674, 559	Gwaneum-dong, Hyeonpung-myeon
2022 local	Jeonnam	Kim Young-rok	Lee Jung-hyun	377, 90	Dongbok-myeon, Jangsan-myeon
2022 local	Chungbuk	Kim Young-hwan	Noh Young-min	319, 158	Yangsan-myeon, Naebuk-myeon

The point is not that historical duplicate pairs never occur. They do. In particular, including the 2014 local election increases the historical duplicate count. But no historical governor contest in the expanded baseline has five repeated pairs within one contest. The historical maximum is three.

A predictable objection is that if three pairs occurred historically, five pairs should not be treated as special. That objection confuses a maximum observation with a tail probability. Three pairs is the upper end observed after pooling 51 historical governor contests. Five pairs exceeds that observed upper end by two additional repeated pairs. Under $N = 393$ and $K = 100,945$, three or more pairs has a probability near 4.23%; five or more pairs has a probability near 0.115%. This paper therefore does not argue that "three pairs are already proof, so five pairs are proof." It accepts three as an observed rare historical upper value and treats five as an event beyond that upper value.

4.2 Empirical Estimate of the Effective Vote-Pair Space

The historical governor dataset contains 1,514,172 total comparison pairs and 15 observed identical vote pairs. If the effective number of possible first-second vote pairs is denoted by K, then the expected number of collisions is roughly:

$$E(C) \approx \frac{N(N-1)}{2K}.$$

Solving backward from the historical governor data gives:

$$\hat{K} \approx \frac{1,514,172}{15} \approx 100,945.$$

This is not a strict structural parameter. Real election data are not generated from identical independent draws; unit size, geography, party support, and turnout matter. But K is an empirical reference value reflecting how often actual Korean governor-election top-two vote pairs collided in official data.

5 Probability Calculation

5.1 Poisson Approximation

Suppose one constituency-level contest contains N in-district early-vote counting units, and each unit produces a top-two vote pair from an effective pair space of size K. Let C be the number of identical vote-pair collisions among distinct unit pairs. For rare collision events, C can be approximated by:

$$C \sim \text{Poisson}(\lambda),$$

with

$$\lambda = \frac{N(N-1)}{2K}.$$

The probability of interest is:

$$P(C \geq 5),$$

which corresponds to at least five identical vote-pair repetitions within the Gwangju-Jeonnam setting.

5.2 Baseline with $N = 393$ and $K = 100,945$

Here N is not the number of already reported rows. It is the number of potential in-district early-vote counting units that could have produced identical pairs within the same contest environment. This means the eup, myeon, and dong level units in the same candidate combination and vote type.

This study does not estimate N from administrative intuition. It counts N directly from the NEC VCCP08 official HTML pages. After parsing all governor counting-unit pages for the five Gwangju districts and the twenty-two Jeonnam cities/counties, and counting rows that are in-district early-vote rows rather than aggregate rows, the output `outputs/nec_2026_gwangju_jeonnam_unit_counts.csv` gives 96 Gwangju units and 297 Jeonnam units, for a total of 393. Therefore the baseline calculation uses $N = 393$.

$K = 100,945$ is also not arbitrary. In the historical governor baseline, there are 1,514,172 actual top-two comparison pairs and 15 observed identical vote pairs. Reversing the rare-collision formula gives K near 100,945.

Using these values:

$$\lambda = \frac{393 \times 392}{2 \times 100,944.8} \approx 0.763.$$

The computed probabilities are:

Threshold	Probability	Approximate odds
$P(C \geq 3)$	0.042260772	about 1 in 23.7
$P(C \geq 4)$	0.007734837	about 1 in 129
$P(C \geq 5)$	0.001148406	about 1 in 871
$P(C \geq 6)$	0.000143224	about 1 in 6,982

Thus the five-pair Gwangju-Jeonnam case is estimated at approximately 0.115% under the baseline model. This does not mean impossibility. It means the event is in the extreme tail under a model already calibrated from real historical Korean election data.

5.3 Sensitivity to the Effective Pair Space

If K is smaller, collisions become easier; if K is larger, collisions become harder. The package therefore reports sensitivity values in `outputs/probability_k_sensitivity.csv`. The central point remains that the observed five-pair cluster is not an ordinary one-pair birthday-problem event. It is an upper-tail event even under a historical-collision calibration that already allows past duplicate pairs.

5.4 Sensitivity to the Number of Counting Units

The paper also checks values around $N = 393$, reported in `outputs/probability_n_sensitivity.csv`. This addresses the objection that a small counting error in the potential unit count could change the conclusion. The exact probability varies with N, but the qualitative conclusion is stable: five repeated pairs is a rare event relative to the historical governor benchmark.

5.5 Nonparametric Resampling Test

The Poisson approximation is useful but simplified. To reduce dependence on a theoretical model, this paper also performs a nonparametric resampling test. The procedure is:

1. Build the pool of actual historical governor top-two vote pairs. 2. Draw 393 units without replacement. 3. Count the number of repeated vote pairs in the draw. 4. Repeat the procedure 200,000 times.

The observed simulation results are:

Threshold	Hits in 200,000 trials	Empirical probability
$C \geq 3$	77	0.000385
$C \geq 4$	4	0.000020
$C \geq 5$	0	0 at simulation resolution

Zero hits in 200,000 trials does not prove mathematical impossibility. It means that the event is below the simulation resolution. Applying the rule of three gives a rough 95% upper bound of about $3/200,000 = 0.000015$, or 0.0015%.

6 Why One Pair and Five Pairs Are Different

The familiar birthday-problem intuition says that duplicate values are not surprising once many comparisons are made. That intuition is correct for "at least one duplicate somewhere." It is not enough to dismiss five repeated top-two vote pairs in the same contest environment.

The difference is the level of aggregation and repetition. One duplicate pair can occur in a large dataset. Five duplicate pairs in the same election, same vote type, same candidate combination, and same regional contest require a different calculation. The relevant probability is not "can a duplicate ever occur?" but "how many duplicate pairs should appear in this defined comparison set?"

The distinction is visible in the numbers. Under the baseline model, $P(C \geq 3)$ is about 4.23%, while $P(C \geq 5)$ is about 0.115%. Both are below the center of the distribution, but five is roughly 37 times rarer than three under the same assumptions. Historical data also show this difference: three pairs occurred as the historical maximum; five did not occur in any historical governor contest in the baseline.

Another key issue is post-search scope. If one searches all elections, all offices, all candidates, all vote types, and all regions, then a rare-looking coincidence will eventually be found. This paper therefore places the main claim on the Gwangju-Jeonnam internal five-pair cluster rather than the entire nationwide set. The Gwangju-Jeonnam cluster is defined by the same office, same top-two candidate pair, same in-district early-vote category, and the same contest environment.

7 Specificity of the Incheon Songdo Case

The Incheon Songdo case is not simply another row to be mechanically added to the Gwangju-Jeonnam cluster. It belongs to a different contest and should be analyzed separately.

The reported final in-district early-vote result is:

Unit	Park Chan-dae final	Yoo Jeong-bok final
Songdo 1-dong	3,030	1,440
Songdo 2-dong	3,030	1,440

The surrounding counts are not identical. The package output `outputs/songdo_official_rows.csv` records different electorate counts, third-candidate counts, invalid votes, and abstentions. In addition, public explanations by the Incheon election-management side stated that the ballot-sorter first-pass results were not identical, and that the final equality emerged after reviewed ballots were hand-counted and added.

This makes the case statistically interesting rather than automatically exculpatory or automatically incriminating. If different first-pass results and different auxiliary quantities lead to identical final top-two results in exactly the two named Songdo units, the review-ballot allocation records become essential.

The reproduction script estimates:

- Yeonsu in-district early-vote units: 15.
- Probability of at least one duplicate pair among the 15 units: about 0.103963%, or about 1 in 962.
- Conditional probability that the specific Songdo 1-dong and Songdo 2-dong pair matches: about 0.0009906%, or about 1 in 100,945.

The Songdo case is therefore best treated as a separate low-probability observation that strengthens the demand for raw records. It should not be merged into the Gwangju-Jeonnam probability calculation without adjustment.

8 Statistical Anomaly Versus Evidence of Misconduct

It is important to separate two propositions:

1. The repeated identical vote-pair pattern is statistically anomalous under the historical benchmark. 2. A specific person or institution committed a specific act of misconduct.

This paper supports the first proposition. It does not, by itself, prove the second. The phrase ”evidence” should therefore be used in two layers. First, identical vote-pair repetition is statistical evidence against the ordinary-randomness explanation. Second, it is not direct evidence identifying a perpetrator, intent, method, or legal violation.

This distinction does not weaken the audit conclusion. It makes the conclusion harder to dismiss. A statistical anomaly need not prove a crime in order to justify raw-data disclosure. Election administration is a public trust system; when an official result produces a rare pattern under a transparent benchmark, the institution must be able to show the records that explain it.

9 Raw Materials Required for Verification

To transform the statistical question into an administrative or legal conclusion, the following records are needed:

1. Original counting statements for all twelve event rows. 2. Ballot-sorter first-pass results for the affected units. 3. Reviewed-ballot counts and candidate allocation records. 4. Manual recount or hand-count logs, if any. 5. Data-entry logs, correction logs, and publication/export logs. 6. Ballot-box movement and chain-of-custody records. 7. Hashable raw exports for the official integrated counting-unit files.

If these records show a coherent unit-by-unit path from first pass to final publication, the anomaly can be explained. If they do not, the statistical anomaly becomes stronger. The present paper cannot resolve that factual question without access to those raw materials.

10 Limitations

The study has several limitations.

First, the 2026 integrated XLSX file comparable to the historical public-data files was not available at the time of writing. The 2026 event rows are rechecked through official NEC HTML pages, but future work should repeat the analysis using the official integrated file or original counting statements.

Second, the Poisson model simplifies the data-generating process. Real voting units are not identical independent draws. Size, geography, turnout, age composition, political preference, and local campaign effects matter. That is why the paper also uses historical actual-pair resampling rather than relying only on a theoretical model.

Third, the early-vote versus election-day vote-share analysis is auxiliary. A systematic early-vote advantage for one party can arise from voter self-selection, party mobilization, demographic differences, pandemic-era voting behavior, and other lawful causes. The analysis is included because the one-directional pattern is statistically notable, not because it independently proves misconduct.

Fourth, the paper does not claim to identify a mechanism. The candidates include possibilities such as ordinary coincidence, administrative correction, display error, data-entry error, review-ballot allocation, or deliberate manipulation. Mechanism identification requires raw records.

11 Early-Vote Versus Election-Day Vote-Share Difference

The same reproduction package also checks National Assembly elections from 2016, 2020, and 2024. For each constituency, the script compares the Democratic candidate's two-party vote share in in-district early voting against the same candidate's two-party vote share on election day.

The output `outputs/early_day_assembly_summary.csv` reports:

Election	Constituencies	Early advantage	Early disadvantage	One-sided binomial p-value
2016 National Assembly	229	118	111	about 0.35
2020 National Assembly	236	236	0	about 9.06×10^{-72}
2024 National Assembly	245	245	0	about 1.77×10^{-74}

11.1 Check Against Official National Assembly Data

This calculation uses official National Assembly counting files and is reproducible through `scripts/analyze_early_day_assembly.py`. The 2016 result does not show a uniform direction. The 2020 and 2024 results do. This makes the pattern difficult to dismiss as ordinary sampling noise alone.

11.2 Connection to the 2026 Local Election

This auxiliary result does not establish the same mechanism as the 2026 identical-pair event. However, it is relevant to the broader audit question because it points to repeated, directionally uniform early-vote differences in recent Korean elections. A strong benign explanation must show why these differences arise from voter self-selection, party mobilization, age structure, occupation, urban/rural composition, or other lawful factors. Without such covariate-adjusted explanation, the pattern remains an additional reason to request raw-data transparency.

12 Expected Objections and Responses

12.1 "With many counting units, identical vote pairs can always appear."

Yes, a single duplicate can appear. The paper does not deny that. The question is whether five repeated top-two pairs in the same defined contest environment are ordinary. The historical benchmark and probability calculations show that five is not the same as one.

12.2 "If 2014 is included, historical duplicates become more common."

Correct. This is why the paper includes 2014 rather than excluding it. The inclusion makes the benchmark more conservative. Even with 2014 included, the historical maximum within one governor contest is three, not five.

12.3 "If three historical pairs are acceptable, five should also be acceptable."

Not under the same probability model. Three or more and five or more have substantially different tail probabilities. Three is a rare but observed historical upper value. Five exceeds that observed upper value and is approximately 37 times rarer than three under the $N = 393, K = 100,945$ Poisson calculation.

12.4 ”Unit sizes and regional preferences invalidate the simple probability calculation.”

They may affect the calculation, which is why the paper does not rely only on a uniform theoretical model. It estimates K from actual Korean governor-election data and performs resampling from historical actual vote pairs. A critic who argues that regional structure fully explains the anomaly should provide a better model and show that it generates five-pair clusters with non-negligible frequency.

12.5 ”The nationwide twelve cases were found after searching, so the probability is overstated.”

This is a serious objection to the nationwide headline. The paper therefore places the main probability claim on the Gwangju-Jeonnam internal cluster rather than the entire nationwide set. The Songdo case is reported separately.

12.6 ”The 2026 data are not yet from an official integrated file.”

This is the most important limitation. The present package rechecks the event rows from official NEC election-statistics HTML, not from a historical-style integrated XLSX file. Therefore the conclusion is not a final legal determination. It is a statistical audit trigger. When the integrated file or original counting statements are released, the event definition should be reapplied.

12.7 ”Early-vote and election-day differences can be caused by voter self-selection.”

Yes. That is why the early-vote analysis is not used as direct proof of misconduct. It is an auxiliary test. A full explanation should include age, occupation, mobility, region, party mobilization, pandemic-era effects, and other variables.

12.8 ”Rare events do sometimes occur.”

Yes. Statistical rarity alone is not impossibility. But public election administration does not require impossibility before audit. A rare official-result pattern can justify raw-data disclosure, especially when the records needed to resolve the question are held by the election-management institution.

12.9 ”How can the claim be falsified?”

The claim is falsifiable. It would be weakened if:

1. Official integrated counting-unit files show that the event rows were misreported or misread. 2. Original counting statements and first-pass machine outputs reconcile the final equal pairs through documented review steps. 3. A broader independently reproduced baseline using the same event definition shows that five-pair clusters commonly occur in comparable contests. 4. A covariate-adjusted model explains the early-vote/election-day pattern without leaving systematic residual anomalies.

13 Raw-Data Audit Decision Criteria

An audit should not be vague. The minimum decision criteria should be:

- Do the official integrated files match the HTML page values?
- Do original counting statements match the published results?

- Do ballot-sorter first-pass results differ from final results, and if so, are all differences explained by review logs?
- Are reviewed ballots allocated by candidate in a way that reproduces the final equality?
- Are data-entry and correction logs consistent with the published timeline?
- Are all twelve event rows traceable from ballot-level or batch-level records to final publication?

If these criteria are met, the anomaly can be administratively explained. If they are not met, the anomaly remains unresolved.

14 Conclusion

The 2026 Gwangju-Jeonnam repeated identical in-district early-vote pairs are not adequately explained by ordinary chance under the tests reported here. The historical governor benchmark contains 51 constituency contests and 1,514,172 actual top-two comparisons, yet no contest reaches five repeated pairs. The Poisson approximation gives about 0.115% for five or more repeated pairs under $N = 393$ and $K = 100,944.8$. The nonparametric resampling test produces zero five-pair trials in 200,000 repetitions.

The Incheon Songdo case adds a separate low-probability exact match whose final equality appears alongside different auxiliary counts. The early-vote/election-day analysis adds a broader directional pattern in recent National Assembly elections. None of these facts alone identifies a culprit or proves a criminal act. Together, however, they create a statistically serious anomaly that cannot be closed by the word "coincidence" without raw evidence.

The proper conclusion is therefore: the current public data do not prove a legal finding of misconduct, but they do establish a strong statistical basis for official raw-data disclosure and independent audit.

15 Data and Code Availability

The reproduction package contains the historical public election files, NEC HTML cache for the 2026 checks, parsing scripts, probability scripts, output CSVs, checksums, the Korean paper, this English paper, and compiled PDFs. The main reproduction command is:

```
python3 scripts/run_all.py
```

The submission package can be rebuilt with:

```
python3 scripts/create_submission_zip.py
```

Package validation can be run with:

```
python3 scripts/validate_package.py
```

16 Research Ethics and Conflict of Interest

This study uses public election data and public reports. It does not process private voter-level data. The author declares no institutional affiliation. The purpose of the study is to define a reproducible statistical anomaly and the raw records needed to resolve it. The paper should not be read as a legal accusation against a specific person without the additional raw administrative records identified above.

Appendix A. Intuitive Meaning of Key Terms

Term	Intuitive meaning
Counting unit	The smallest official result row used for comparison, such as an eup, myeon, or dong in-district early-vote row
In-district early voting	Early votes cast by voters within their registered district
Identical vote pair	Two counting units have the same first-candidate count and same second-candidate count
Effective pair space K	The empirically inferred range of possible top-two vote pairs
Tail probability	The probability of observing at least the reported level of repetition
Poisson approximation	A convenient approximation for rare collision counts
Nonparametric resampling	A simulation that samples from actual historical vote pairs rather than a purely theoretical distribution
Audit trigger	A signal strong enough to require raw-data disclosure and independent verification, without itself proving misconduct

Appendix B. Early Voting and the Law of Large Numbers

Public discussions sometimes claim that the ”law of large numbers” should force early-vote and election-day vote shares to be nearly identical. That is not generally correct. The law of large numbers applies when samples are drawn from the same underlying distribution or from comparable random processes. Early voters and election-day voters are not random samples from a single identical population; they may differ by age, occupation, geography, political interest, party mobilization, and other factors.

Therefore a single early-vote advantage is not suspicious by itself. The more relevant question is whether the direction and size of the difference repeat so uniformly that voter self-selection alone becomes insufficient. In the official National Assembly data checked here, 2020 and 2024 show Democratic early-vote two-party advantages in all analyzable constituencies. Under a simple equal-direction benchmark, the one-sided binomial probabilities are extremely small. This is not used as direct proof of a specific misconduct mechanism, but it supports the need for a covariate-adjusted explanation or raw-data audit.

References

- National Election Commission, election-statistics system, counting-unit result pages.
- Public Data Portal, National Election Commission vote-counting data files for past elections.
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- Public Data Portal, 2020 National Assembly election counting data.
- Public Data Portal, 2022 nationwide local election counting data.
- Public Data Portal, 2024 National Assembly election counting data.

- News1/Daum, June 6, 2026, report on identical vote counts in Songdo 1-dong and Songdo 2-dong.
- Hankyoreh, June 8, 2026, report on the Songdo identical-count controversy and election-management explanation.
- TV Chosun, June 8, 2026, report on identical in-district early-vote counts in twelve places nationwide.

Reproduction Outputs

Key files in the package include:

- `outputs/dataset_counts.csv`
- `outputs/duplicate_summary.csv`
- `outputs/governor_actual_top2_summary.csv`
- `outputs/governor_actual_top2_by_contest.csv`
- `outputs/governor_actual_top2_duplicates.csv`
- `outputs/governor_bootstrap_summary.csv`
- `outputs/governor_bootstrap_histogram.csv`
- `outputs/nec_2026_gwangju_jeonnam_unit_counts.csv`
- `outputs/nec_2026_gwangju_jeonnam_units.csv`
- `outputs/nec_2026_gwangju_jeonnam_unit_summary.json`
- `outputs/nec_2026_reported_duplicate_cases.csv`
- `outputs/nec_2026_reported_duplicate_pairs.csv`
- `outputs/songdo_probability_summary.csv`
- `outputs/songdo_official_rows.csv`
- `outputs/probability_core.csv`
- `outputs/probability_k_sensitivity.csv`
- `outputs/probability_n_sensitivity.csv`
- `outputs/early_day_assembly_summary.csv`
- `outputs/early_day_assembly_twoparty.csv`
- `outputs/checksums_sha256.csv`